

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

NVGo: Mobile Application for Efficient Communication and Service Management in Nueva Valencia

In partial fulfillment for the requirements in Capstone 1 and Research Subject

Sayo, Clark Eldine- BSIT 3-6

Pami, John Mark-BSIT3-6

Tamayo, Oyo Frederick - BSIT3-6

Viñarta, Jude Ryan - BSIT3-7

Rodriguez, Adrian - BSIT3-7

System Development Track

Second Semester

2025-26

CHAPTER 1

INTRODUCTION

1. 1 Background of the Study

The rapid growth of modern technology in the Philippines has greatly changed the way people communicate with each other. This makes socialization convenient for people. However, despite this convenience, not every island in the Philippines can catch up to the modern technological advancements due to their environmental challenges and remote location.

Hon. Emmanuel C. Galila is the mayor of the municipality of Nueva Valencia, Guimaras. They have been handled by the DICT recently. They deployed Starlink internet connections all throughout Nueva Valencia, especially in their remote islands. They can now connect with their residents online.

But despite being given accessibility in communication, there are things that are limited. They are vulnerable to disinformation. They want to make sure that the information that they will send and receive is secure and legitimate. With the internet being available in the area, you cannot expect a safe exchange of information.

To resolve these challenges, the municipality wants us to create a mobile application that can verify reports and respond with the right equipment and better information dissemination. We also agreed to add some features that can help important transactions to be convenient for residents that are unable to visit the municipality personally. So we proposed NVGo, a mobile application for efficient communication and service management in Nueva Valencia. With Starlink being deployed, this will help them enhance their information dissemination and precise service management.

1.2 Statement of the problem

Despite the availability of internet connection in remote areas, several issues remain unresolved. This study seeks to address the following problems:

1. Insecure exchange of communication from both parties.
2. Illegitimate report on infrastructure damage.
3. Misrouted emergency calls lead to improper response and delays.
4. Fake profiles can be a threat and can cause false alarms.

1.3 Objectives of the study

1.3.1 General Objectives

The general objective of this study is to design and develop a mobile application that is reliable and convenient, improves citizen's communication, proper emergency calls, and legit profiles.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. To design a user-friendly interface for information dissemination.
2. To implement proper input reports and emergency response.
3. To test the app's usability among citizens in remote areas.
4. To evaluate the effectiveness of a mobile app in improving convenience.
5. To improve municipality services.

1.4 Scope and Delimitation

The study focuses on the development of a mobile application that uses OOP in python language using Flask as the framework. This system will start from finishing the CRUD system first then transform the frontend into REACT for real-time features like notification and live updates, then convert it to mobile. The system will include user registration, profile creation, administrative dashboard, and real-time updates.

The study will not cover large-scale data processing, or advance data analysis. The system is developed as a prototype to demonstrate the feasibility and functionality of the proposed solution. It focuses on core features and may be enhanced further for real-world deployment.

1.5 Significance of the study

The proposed system will benefit the following:

- **Residents of Nueva Valencia** – for easy complaint submission, incident reporting, and access to

announcements.

- **Municipality Officials** – for efficient management and monitoring of services and reports.
- **Municipality of Nueva Valencia** – for improved service delivery and operational efficiency.
- **Future Researchers** – as a reference for similar mobile-based systems.
- **Proponents** – for enhancing development and technical skills.

1.6 Definition of Terms

- **DICT** - DICT stands for the Department of Information and Communications Technology, the Philippine government agency responsible for the country's ICT development and digital transformation.
- **Starlink** - Starlink is a satellite internet service operated by SpaceX that provides high-speed, low-latency broadband globally, even in remote areas.
- **Information Dissemination** - is the process of distributing and sharing information to a targeted audience through various channels, ensuring clarity and accessibility.
- **OOP** - is a programming paradigm that uses the concept of classes and objects to structure software programs into reusable and modular pieces of code.
- **REACT** - is a JavaScript library created by Facebook for building user interfaces, particularly for single-page applications. It allows developers to create reusable UI components, which can manage their own state and be composed to build complex user interfaces.
- **Flask** - is a lightweight web application framework written in Python.

CHAPTER 2

REVIEW RELATED LITERATURE AND STUDIES

This chapter presents relevant foreign and local studies, related literature, and a synthesis of findings that provide the foundation for the present study on NV Go: Mobile App for Efficient Communication and Service Management in Nueva Valencia.

2.1 Foreign Studies

According to Janssen and Estevez (2016), mobile governance applications in Europe and Asia have significantly improved service delivery by reducing bureaucratic delays and enhancing transparency. Their study emphasized that mobile-first platforms are particularly effective in municipalities with limited resources, as they provide cost-efficient communication channels between citizens and government offices.

Similarly, Tapscott (2019) highlighted that mobile applications foster participatory governance by

enabling real-time feedback, emergency alerts, and community engagement. In India, Sharma et al. (2020) found that mobile apps used for municipal service management increased citizen satisfaction and reduced response times for complaints and requests. These findings demonstrate the global relevance of mobile platforms in strengthening governance and service management.

2.2 Local Studies

In the Philippine context, Alampay (2018) examined ICT adoption in local government units (LGUs) and found that mobile applications improved responsiveness and accountability. His study emphasized that mobile-first solutions are crucial in rural areas where smartphones are more accessible than desktop computers.

David et al. (2020) studied mobile communication platforms in Visayas municipalities, noting their effectiveness in disaster preparedness and community coordination. Reyes (2021) further emphasized that integrating mobile service management systems enhances reliability and usability, particularly when designed with user-centered principles.

Although direct documentation of NV Go in Nueva Valencia is limited, these local studies provide evidence that similar mobile applications have been successfully implemented in Philippine municipalities to improve communication and service delivery.

2.3 Related Literature

Heeks (2017) explained that e-governance initiatives worldwide rely on ICT tools to deliver services more effectively, highlighting mobile apps as key instruments in bridging communication gaps. Pressman (2019) stressed the importance of structured software engineering methodologies in ensuring system quality, which applies directly to mobile governance platforms.

Soriano (2019) discussed challenges in digital transformation in rural municipalities, including limited internet connectivity, digital literacy gaps, and funding constraints. These challenges are critical considerations for the sustainability of mobile applications like NV Go.

Kim and Lee (2020) emphasized that user-centered design and continuous feedback mechanisms are essential for long-term success in mobile governance systems, ensuring citizen trust and adoption.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies demonstrate that mobile applications significantly enhance communication, service management, and citizen engagement in local governance. Foreign studies highlight global successes in mobile governance platforms, while local studies confirm their relevance

and effectiveness in Philippine municipalities.

However, limited research focuses specifically on Nueva Valencia's NV Go app. This study addresses that gap by situating NV Go within the broader context of mobile governance and ICT adoption in rural municipalities, thereby contributing to the literature on digital transformation in local governance.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study will use a **Developmental Research Design**, focusing on the design, development, and evaluation of a system intended for the residents and local government of Nueva Valencia. This system aims to streamline business permit processing and promote tourist destinations within the municipality.

3.2 System Development Methodology

This study will use Agile Development Model, The following phases will be implemented:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype
6. Improvement (Iteration)



3.3 System Architecture

The system will follow a Client-Server Architecture, where users such as residents, tourists, and administrators will access the system through a web or mobile interface. This server is responsible for processing business permit applications, managing news and announcements, handling and database operations. This setup ensures efficient data handling and accessibility for all users.

3.4 Tools and Technologies Used

- The system will be developed using the following tools and technologies:
- Frontend : HTML,

- Backend: Python Flask
- Machine Learning: Python
- Database: MySQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

- Google survey form and questionnaires
- Interviews
- Observation

3.6 Testing Procedures

- Unit Testing - To ensure that every function operates as intended, individual system components are tested.
- System Testing - assesses the entire system to make sure it satisfies the requirements.
- User Acceptance Testing (UAT) - Performed with real users to confirm functionality and usability in a practical setting.

3.7 Evaluation Criteria

The develop system will be evaluated based on the following criteria:

- Functionality - The degree to which the system carries out its intended functions accurately.
- Usability - The ease of use and user-friendliness of the system interface.
- Efficiency - The system's ability to perform tasks quickly and with minimal resource usage.
- Security - The system's capability to protect user data and prevent unauthorized access.
- Reliability - The consistency of system performance under various conditions

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

The Data Flow Diagram illustrates how data moves through the system and how different components interact with each other.

Process Description:

- Users submit business permit applications through the system.

- The system processes the submitted data and stores it in the database.
- Administrators review the applications and either approve or reject them.
- The system also allows administrators to post announcements, which are then accessed by users.

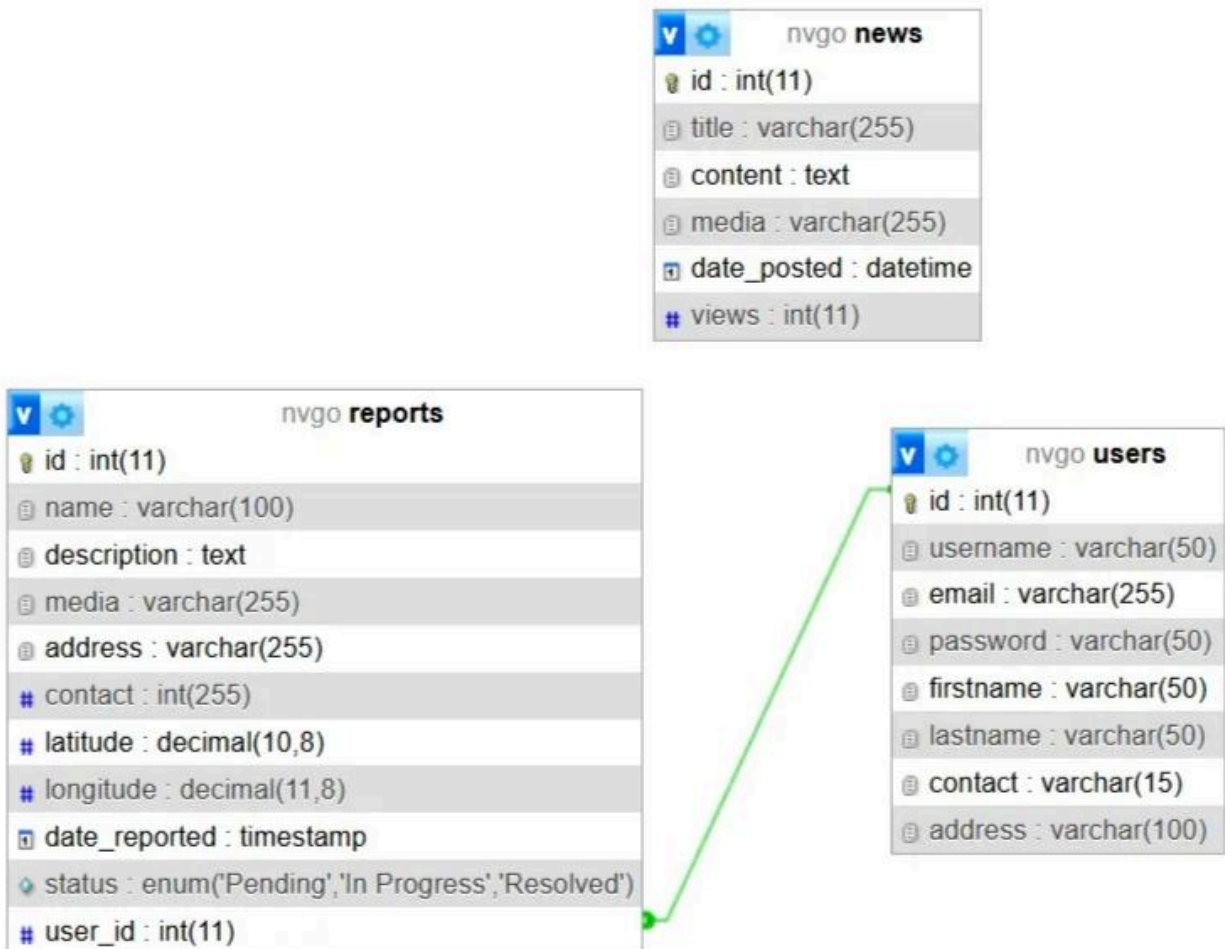
Flow Representation:

User → System → Database

This diagram provides a high-level overview of system operations and data movement.

Entity Relationship Diagram (ERD)

Entities:



Relationships:

- User Submit a Report
- Admin Verifies the Report

- Admin call the Right Response
- Admin Update the Report
- User Receives the Report Update

Use Case Diagram

Actors:

- User
- Administrator

Use Cases:

- Register Account
- Login Account
- Apply for Business Permit
- Track Application Status
- View Announcements
- Approve/Reject Applications (Admin)
- Post Announcements (Admin)